

GRINTEROL® 250 mg hard capsules

Ursodeoxycholic acid (250 mg)

Product Description

GRINTEROL® 250 mg hard capsules are white hard gelatin capsules size 0 containing white or almost white powder.

- The active ingredient is ursodeoxycholic acid (UDCA).
- The other ingredients are: maize starch, silicon dioxide (E 551), magnesium stearate (E 470B), hard gelatin capsule (body and cap composition): titanium dioxide (E 171), gelatin (E 441).

Pharmacodynamic properties

Pharmacotherapeutic group: bile therapy, bile acid and derivatives, ATC code: A05AA02

Bile acids are the most important components of bile and play a role in stimulating bile production. Bile acids are also important to keep cholesterol dissolved in bile. In healthy individuals, the ratio between concentrations of cholesterol and bile acids in the bile is such that cholesterol is kept dissolved for most of the day. Thus, no gallstones can form (bile is non-lithogenic). In patients with cholesterol stones in the gallbladder, this ratio is altered, and the bile is supersaturated with cholesterol (bile is lithogenic). After some time, this may cause precipitation of cholesterol crystals and the formation of gallstones. This medicinal product can convert lithogenic bile into non-lithogenic bile and also gradually dissolve cholesterol gallstones.

Studies into the effect of this medicinal product on cholestasis in patients with impaired biliary drainage and on clinical symptoms in patients with biliary cirrhosis and cystic fibrosis have shown a rapid decline in cholestatic symptoms in the blood (as measured by increased levels of alkaline phosphatase (AP), gamma-GT and bilirubin) and pruritus, as well as decreased fatigue in most patients.

Paediatric population

Cystic fibrosis

From clinical reports long-term experience up to 10 years and more is available with UDCA treatment in paediatric patients with hepatobiliary disorders as a result of cystic fibrosis (CFAHD). It has been demonstrated that treatment with UDCA can decrease bile duct proliferation, halt progression of histological damage and even reverse hepatobiliary changes, when it is given at early stage of CFAHD. Treatment with UDCA should be started as soon as the diagnosis of CFAHD is made, in order to optimise treatment effectiveness.

Pharmacokinetic properties

Absorption

About 60-80 % of orally administered ursodeoxycholic acid is rapidly absorbed in jejunum and in upper ileum by passive diffusion and in terminal ileum by active transport.

Distribution

Following absorption, ursodeoxycholic acid passes into the liver (there is a considerable "first-pass-effect"), where it is conjugated with glycine or taurine and subsequently excreted into the biliary tract. Only a small proportion of ursodeoxycholic acid is found in the systemic circulation and this is renally excreted.

After repeated dosing, the ursodeoxycholic acid concentration in bile reaches steady state after about 3 weeks: however, the total concentration of ursodeoxycholic acid is never any higher than about 60 % of the total bile acid concentration in bile, even at high dosages.

Biotransformation and elimination

With the exception of conjugation, ursodeoxycholic acid is not metabolised. However, a small amount of orally administered ursodeoxycholic acid undergoes bacterial conversion to 7-keto-lithocholic acid or lithocholic acid after each enterohepatic circulation, while bacterial deconjugation in the duodenum also takes place.

Since ursodeoxycholic acid, 7-keto-lithocholic acid and lithocholic acid are relatively poorly soluble in water, a large amount is excreted via the bile in the faeces. Absorbed ursodeoxycholic acid is reconstituted by the liver; 80 % of the lithocholic acid produced in the duodenum is excreted in the faeces, but the remaining 20 % is sulphated following absorption by the liver to insoluble lithocholyl conjugates, which are then excreted via the bile and faeces. Absorbed 7-keto lithocholic acid is reduced to chenodeoxycholic acid in the liver.

Lithocholic acid can cause cholestatic liver damage when the liver is not able to sulphate lithocholic acid. Although a reduced capacity to sulphate lithocholic acid in the liver has been found in some patients, there is provisionally no clinical evidence to suggest that cholestatic liver damage can be associated with ursodeoxycholic acid therapy.

Upon discontinuation of therapy with ursodeoxycholic acid, the concentration of ursodeoxycholic acid in bile after 1 week rapidly declines to 5-10 % of the steady-state concentration.

The biological half-life of ursodeoxycholic acid is approximately 3.5-5.8 days.

Indication

- Dissolution of radiolucent cholesterol gallstones in the gallbladder less than 15 mm in diameter. The gallstones must be radiolucent, and the gallbladder function must be intact; dissolution of gallstones with or without preceding extracorporeal shock wave lithotripsy.
- Cholestatic liver disease (e.g. compensated primary biliary cirrhosis, cholestasis of pregnancy).

Recommended Dosage

For dissolution of cholesterol gallstones (approx. 10mg per kg bodyweight daily):

		Time of administration
Body weight	Daily dose	In the evening
Up to 60 kg	2 capsules	2
61 – 80 kg	3 capsules	3
81 – 100 kg	4 capsules	4
Over 100 kg	5 capsules	5

For the treatment of cholestatic liver disease:

The daily dose depends on bodyweight and is approx. 2 to 6 capsules (approx. 10 to 15 mg ursodeoxycholic acid per kg bodyweight).

		Time of administration		
Body weight	Daily dose	Morning	Midday	Evening
34 – 50 kg	2 capsules	1	-	1
51 – 65 kg	3 capsules	1	1	1
66 – 85 kg	4 capsules	1	1	2
86 – 110 kg	5 capsules	1	2	2
Over 110 kg	6 capsules	2	2	2

Mode and duration of Administration

Swallow the capsules whole with some liquid. Take the capsules regularly.

Dissolution of gallstones generally takes from 6 to 24 months to dissolve. If the gallstones have not become smaller after 12 months, do not continue treatment. The results of therapy should be monitored by ultrasonography or X-ray every 6 months.

Contraindications

- Hypersensitivity to the active substance, bile acids or to any of the other ingredients of GRINTEROL® capsules.
- Acute inflammation of the gall bladder or biliary tract.
- Occlusion of biliary tract (occlusion of the common bile duct or a cystic duct).
- Repeated biliary colic.
- Radio-opaque calcified gallstones.
- Impaired contractility of the gall bladder.

Paediatric population:

- Unsuccessful portoenterostomy or missing recovery of normal bile flow in children with biliary atresia.

Warnings and Precautions

Ursodeoxycholic acid should be used under medical supervision.

During the first 3 months of treatment, liver function parameters AST (SGOT), ALT (SGPT) and γ -GT should be monitored by the physician every 4 weeks, and thereafter every 3 months. As well as allowing for differentiation between responsive and non-responsive patients treated for PBC, this monitoring also enables early detection of potential deterioration in liver function, particularly in patients with advanced stage PBC.

When used for the dissolution of cholesterol gallstones

In order to assess the therapeutic progression of gallstone dissolution and to promptly detect any calcification of the gallstones, the gall bladder should be visualised (oral cholecystography) 6 to 10 months after the beginning of treatment, depending on stone size, with overview and occlusions in standing and supine positions (ultrasound control).

If the gall bladder cannot be visualised on X-ray images, or in cases of calcified gallstones, impaired contractility of the gallbladder or frequent episodes of biliary colic, treatment with this medicinal product must be discontinued.

Female patients taking this medicinal product to dissolve gallstones should use an effective non-hormonal contraceptive, as hormonal contraceptives may promote the formation of gallstones.

When used to treat patients with advanced stage of primary biliary cirrhosis (PBC)

In very rare cases decompensation of hepatic cirrhosis has been observed, which partially regressed after the treatment was discontinued.

In patients with PBC, in rare cases the clinical symptoms may worsen at the beginning of treatment, e.g. the itching may increase. In this case, the dosage of this medicinal product should be reduced to one 250 mg capsule daily and subsequently should be gradually increased to the recommended dose as described in section “*Recommended Dosage*”.

If diarrhoea occurs, the dosage must be reduced and, in the event of persistent diarrhoea, treatment must be discontinued.

Effects on ability to drive and use machines

UDCA has no or negligible influence on the ability to drive and use machines.

Interactions with Other Medicinal Products

UDCA should not be administered concomitantly with colestyramine, colestipol, charcoal or antacids containing aluminium hydroxide and/or smectite (aluminium oxide), because these preparations bind UDCA in the intestine and thereby inhibit its absorption and efficacy. If the use of any such medicine is needed it must be taken at least two hours before or after UDCA.

UDCA can affect the absorption of ciclosporin from the intestine. In patients treated with ciclosporin, blood concentrations of this substance should therefore be checked by the physician and the ciclosporin dose adjusted if necessary.

Due to the effect of ursodeoxycholic acid on the secretion of bile acids there is a theoretical possibility that the absorption of other lipophilic substances could be affected.

In isolated cases UDCA can reduce the absorption of ciprofloxacin.

In a clinical study in healthy volunteers, concomitant use of UDCA (500 mg/day) and rosuvastatin (20 mg/day) resulted in slightly elevated plasma levels of rosuvastatin. The clinical relevance of this interaction also with regard to other statins is unknown.

This medicinal product reduces the peak plasma concentration ($C_{\max}$) and the area under the curve (AUC) of the calcium antagonist nitrendipine in healthy volunteers. Close monitoring of the outcome of concomitant use of nitrendipine and UDCA is recommended. It may be necessary to increase the dosage of nitrendipine. An interaction with dapsone has also been reported, with a reduction in its therapeutic effect. These observations, together with in vitro data, might indicate that UDCA can induce cytochrome P450 3A enzymes. However, induction has not been observed in a well-designed interaction study with budesonide, which is a known cytochrome P450 3A substrate.

Estrogenic hormones and blood cholesterol-lowering agents such as clofibrate increase hepatic cholesterol secretion and may thereby stimulate formation of gallstones; this effect is counteractive in the use of this medicinal product for gallstone dissolution.

Pregnancy and Lactation

Pregnancy

There are no or limited amounts of data from the use of UDCA in pregnant women. Studies in animals have shown reproductive toxicity during the early phase of gestation. Therefore, this medicinal product should not be used during pregnancy unless clearly necessary.

Women of childbearing potential

Women of childbearing potential may only be treated with UDCA if they use reliable contraception: non-hormonal or oral contraception with a low-oestrogen dosage is recommended. However, in patients taking GRINTEROL[®] capsules for dissolution of gallstones, effective non-hormonal contraception should be used, as hormonal oral contraceptives may increase formation of gallstones.

The possibility of a pregnancy must be excluded before beginning treatment.

Breast-feeding

According to few documented cases of breastfeeding women, the amount of UDCA in milk was very low and probably no adverse reactions are to be expected in breastfed infants.

Fertility

Animal studies did not show an influence of UDCA on fertility. Human data on fertility effects following treatment with UDCA are not available.

Undesirable Effects

The evaluation of undesirable effects is based on the following frequency data:

Very common ($\geq 1/10$)
Common ($\geq 1/100$ to $< 1/10$)
Uncommon ($\geq 1/1\ 000$ to $< 1/100$)
Rare ($\geq 1/10\ 000$ to $< 1/1\ 000$)
Very rare ($< 1/10\ 000$),
Not known (cannot be estimated from the available data).

Gastrointestinal disorders

Common: diarrhoea, pasty stools.

Very rare: severe right upper abdominal pain during the treatment of PBC.

Hepatobiliary disorders

Very rare: calcification of gallstones; decompensation of liver cirrhosis (during the treatment of advanced stages of PBC), which partially regressed upon discontinuation of treatment.

Skin and subcutaneous tissue disorders

Very rare: urticaria.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. You may report side effects or adverse drug reactions directly to the National Centre for Adverse Drug Reaction Monitoring by calling Tel: 03-78835550, or visiting the website: npra.moh.gov.my (*Public* ☐ *Consumers* ☐ *Reporting Side Effects to Medicines (ConSERF) or Vaccines (AEFI)*)

Overdose and Treatment

Diarrhoea may occur in the event of overdose. In general, other symptoms of overdose are unlikely, as the absorption of this medicinal product decreases with increasing dose and therefore more is excreted with the faeces.

No specific counter-measures are necessary and the consequences of diarrhoea should be treated symptomatically with restoration of the fluid and electrolyte balance.

Additional information on special populations

Long-term use of high-dose UDCA therapy (28-30 mg/kg/day) in patients with primary sclerosing cholangitis (off-label use) was associated with an increase of severe adverse reactions.

Storage Conditions

Store in the original package at temperature not exceeding 30 °C.

Keep out of reach of children.

Shelf life

4 years.

Dosage Forms and Packaging Available

Each capsule contains 250 mg of ursodeoxycholic acid.

10 capsules per blister. Pack sizes: 50, 60 or 100 capsules.

Name and Address of Manufacturer

JSC GRINDEKS.

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